

|       |     |                                                                                                                                                                                                                                           |    |     |
|-------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
| 1     | (a) | scalar: magnitude only                                                                                                                                                                                                                    | B1 | [2] |
|       |     | vector: magnitude and direction ( <i>allow scalar with direction</i> )<br>( <i>allow 1 mark for scalar has no direction, vector has direction</i> )                                                                                       | B1 |     |
|       | (b) | diagram has correct shape                                                                                                                                                                                                                 | M1 | [4] |
|       |     | with arrows in correct directions                                                                                                                                                                                                         | A1 |     |
|       |     | resultant = $13.2 \pm 0.2$ N ( <i>allow 2 sig. fig</i> )                                                                                                                                                                                  | A2 |     |
|       |     | ( <i>for <math>12.8 \rightarrow 13.0</math> and <math>13.4 \rightarrow 13.6</math>, allow 1 mark</i> )<br>( <i>calculated answer with a correct sketch, allow max 4 marks</i> )<br>( <i>calculated answer with no sketch – no marks</i> ) |    |     |
| Total |     |                                                                                                                                                                                                                                           |    | [6] |
| 2     | (a) | (i) $\lambda = 0.6$ m                                                                                                                                                                                                                     | B1 |     |
|       |     | (ii) frequency ( $= v/\lambda$ ) = $330/0.60$                                                                                                                                                                                             | C1 |     |